

A study on the classification method of motor imagery EEG signals based on band filter bank common space pattern and continuous wavelet transforms

Qixuan Cui*

School of Electronics and Information
Xi'an Polytechnic University
Xi'an, China

*Corresponding author: e-mail: m17792538175@163.com

Xiaodan Zhang

School of Electronics and Information
Xi'an Polytechnic University
Xi'an, China

e-mail: zhangxiaodan@xpu.edu.cn

Kemeng Xu

School of Electronics and Information
Xi'an Polytechnic University
Xi'an, China
e-mail: xvkemeng@126.com

Shuyi Wang

School of Electronics and Information
Xi'an Polytechnic University
Xi'an, China
e-mail: wang_sy@163.com

Abstract: EEG signal feature extraction link will affect the accuracy of MI, this paper is optimized for feature extraction to improve the classification accuracy. A feature extraction model (Band Filter Bank Common Spatial Pattern-Continuous wavelet transform, CCWT) is proposed to address the problem that the airspace features extracted by Common Spatial Pattern (CSP) are relatively homogeneous. The model is based on the band filter bank co-space pattern fusion Continuous wavelet transform (CWT) algorithm. The original data are first divided into multiple frequency bands, and the spatial domain features are extracted by CSP, and then the features are processed by continuous wavelet transform, so that the features contain time-frequency domain information, and the final feature matrix is obtained. The study validates the proposed method CCWT on the public dataset BCI Competition IV 2a dataset with an average accuracy of 86.713% and an average Kappa of 0.8231.

Keywords-Band Filter Bank Common Spatial Pattern; Continuous wavelet transform; Motion imagery

I. INTRODUCTION

Brain-computer interface (BCI) is a communication technology designed to establish the transfer of information between the human brain and a computer, opening up a new way for the brain to interact with the outside world [1]. This technique is mainly used in areas such as rehabilitation therapy [2][3]. Motor Imagery (MI), as one of the important applications of brain-computer interfaces, decodes the EEG signals generated when a user performs a limb-imagery movement for intent recognition. The user's motor activity is simulated internally without the accompaniment of visible physical movement [4].

In the field of motor imagery EEG signal research, feature extraction is an important aspect that affects classification accuracy. Literature [5] proposed a band optimization algorithm based on Common Spatial Pattern (CSP) and compared it with three CSP algorithms and achieved high classification accuracy. Literature [6] compared the performance of machine learning

and deep learning classification methods. In terms of machine learning, the study used CSP for feature extraction, and then a Linear Discriminant Analysis (LDA) model was used for binary classification of MI tasks. In terms of deep learning, the study constructed a convolutional neural network (CNN) model of the raw EEG signals. The results of the study showed that the CNN model improved the classification accuracy by 2.37 - 28.28% over machine learning. A method combining continuous wavelet transform (CWT) and simplified convolutional neural network (SCNN) was proposed in the literature [7] to improve the recognition rate of MI-EEG signals. The study obtained a mean classification accuracy of 83.2% and a mean kappa of 0.651 on 9 subjects from BCI Competition IV Dataset 2b. Literature [8] first constructed multi-scale sub-signals simultaneously based on time windows and spectral bands, and extracts CSP features from each sub-signal before classifying them using a Multi-Center SCA (MCSCA) classifier. The study obtained good results at BCI Competition IV Dataset 2a.

The above studies have achieved good results, but the extracted features are relatively homogeneous and cannot fully characterize the motor imagery EEG signals. In this study, we propose a feature extraction method Band Filter Bank Common Spatial Pattern-Continuous wavelet transform (CCWT) that can make the features contain both spatial and time-frequency information.

II. MATERIALS AND METHODS

In this study, nine subjects from the BCI Competition IV 2a dataset were investigated in terms of data preprocessing, extraction of time-frequency and space domain fusion features and classification.

A. Data sources

The BCI Competition IV 2a dataset consisted of 9 subjects, and the EEG signals collected were 25 channels of EEG signals. The dataset samples the signal at 250 Hz and is band-pass filtered between 0.5 Hz and 100 Hz. Subjects completed a four-

category motor imagery task lasting four seconds for the left hand, right hand, feet, and tongue in response to a cue. Each subject had two sets of data used for testing and training, and each set of data contained nine structs, the last six of which corresponded to the six runs of this session. The size of EEG data and labels are 96735×25 and 48×1 , respectively, and the EEG data are divided into 48 segments corresponding to the labels by a trial of size 48×1 .

B. Data pre-processing

In the preprocessing process, this study transforms the raw 2D EEG data into 3D signals. Figure 1 shows the data dimension transformation process, which mainly contains 2 steps:

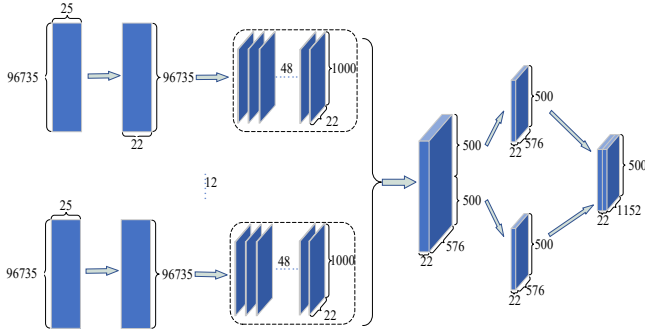


Figure 1 Data dimension transformation

1) Take 2-6 seconds of data from the Competition IV 2a dataset, combine the 2 sets of data for each subject, and randomly divide the training and test sets in subsequent model training.

2) The 1000 samples were divided into two and spliced according to the third dimension trails to get twice the original number of samples.

C. Band Filter Bank Common Spatial Pattern

In this study, the preprocessed data were divided into eight frequency bands of 4-7 Hz, 7-12 Hz, 12-16 Hz, 16-22 Hz, 22-28 Hz, 28-32 Hz, 32-36 Hz, and 36-40 Hz. CSP features are extracted in each band to enhance interclass variation and reduce intraclass variation. The BCI Competition IV 2a dataset is a 4-categorization task, which does not allow for the direct extraction of CSP features. In this study, a binary classifier is trained for each category, and all the other categories are merged into a single class, converting the four-categorization problem into four binary problems.

For CSP feature extraction, the intra-class covariance matrix is first calculated:

$$C_1 = \frac{1}{N_1} \sum_{i=1}^{N_1} x_i x_i^T \quad (1)$$

$$C_2 = \frac{1}{N_2} \sum_{i=1}^{N_2} y_i y_i^T \quad (2)$$

where N_1 is the number of samples in the positive category, x_i is the multichannel EEG signal of the i th sample, C_1 is the covariance matrix of the positive category. N_2 is the number of samples in the negative category, y_i is the multichannel EEG signal of the i th sample. C_2 is the covariance matrix of the negative category.

Next, the total covariance matrix is calculated:

$$C = C_1 + C_2 \quad (3)$$

Then, the generalized eigenvalue problem is solved:

$$C_1 W = \lambda W \quad (4)$$

where λ is the eigenvalue and W is the eigenvector. The eigenvalues λ are sorted in order of size and the eigenvectors corresponding to the largest and smallest few eigenvalues are selected as the optimal spatial filters. These eigenvectors maximize the variance ratio between the two types of signals.

Finally, the features are extracted:

$$z = W^T x \quad (5)$$

where x is the original multichannel EEG signal and z is the filtered feature vector.

D. continuous wavelet transform

The continuous wavelet transform provides a time-frequency representation of a signal by decomposing it into wavelets of different scales. For the analysis of electroencephalogram (EEG) signals with a high degree of non-stationarity, the continuous wavelet transform is highly adaptable and can effectively deal with mutations, noise, and other non-linear features in EEG signals. These features are essential for understanding brain function and diagnosing various neurological disorders.

The continuous wavelet transform function is as follows:

$$\omega_s(\alpha, \beta) = \frac{1}{\sqrt{\alpha}} \int_{-\infty}^{\infty} s(t) \varphi\left(\frac{t-\beta}{\alpha}\right) dt \quad (6)$$

where $\omega_s(\alpha, \beta)$ denotes the local features of the signal at scale α and position β . $s(t)$ is the input motor imagery EEG signal feature matrix, α is the scale of the wavelet transform controlling the temporal window width and frequency resolution of the wavelet. β is the temporal translation parameter of the wavelet, denoting the position of the wavelet on the time axis. φ is the wavelet basis function.

III. RESULTS & DISCUSSION

This study tested the proposed model on nine subjects from the Competition IV 2a dataset and studied and analyzed the model accuracy and kappa values.

Figure 2 shows the iterative curve for subject 9:

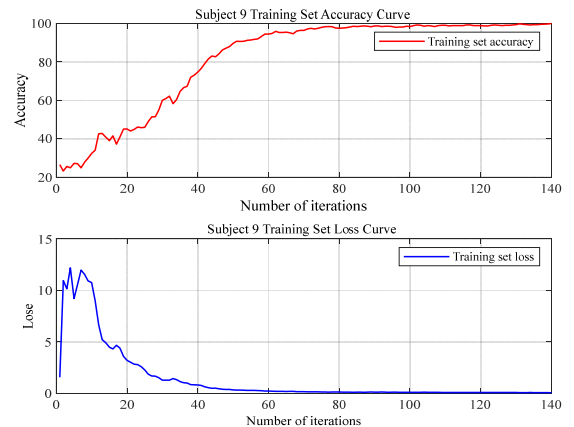


Figure 2 Iteration curve

Figure 3 shows the confusion matrix for the subject 9 test set:

This study randomly divided the 1152 samples of subject 9 into training and test sets according to 900:252. The test set shown in Figure 3 has 65 samples of 1-class, 65 samples of 2-class, 56 samples of 3-class, and 66 samples of 4-class. Five of the 1-class samples were misclassified as class 2 and two were misclassified as class 3. The average classification accuracy of the 1-class samples was 89.2%. Four of the 2-class samples were misclassified as class 1 and two as class 4. The average classification accuracy for the 2-class samples was 90.8%. Two of the 3-category samples were misclassified as class 1. The average classification accuracy of the 3-category samples was 96.4%. Two of the 4-class samples were misclassified as class 1, four as class 2 and two as class 3. The average classification accuracy of the 4-class samples was 87.9%.

1	58	5	2		89.2%	10.8%
2	4	59		2	90.8%	9.2%
3	2		54		96.4%	3.6%
4	2	4	2	58	87.9%	12.1%

	87.9%	86.8%	93.1%	96.7%
	12.1%	13.2%	6.9%	3.3%
	1	2	3	4

Figure 3 Confusion matrix

In this study, all nine subjects were tested five times and the average accuracy and kappa values are shown in Table 1:

Table 1 Experimental results

Subject	Accuracy(%)	Kappa
1	83.7500	0.7839
2	87.6984	0.8347
3	88.1250	0.8418
4	88.0952	0.8439
5	86.5079	0.8203
6	85.7143	0.8100
7	87.3016	0.8308
8	89.6825	0.8625
9	90.8730	0.8784
Ave	87.5275	0.8340

On the nine subjects of the Competition IV 2a dataset, the method proposed in this study had the lowest average accuracy of 83.7500% on subject 1 and the highest average accuracy of 90.8730% on subject 9. The average accuracy of the dataset reached 87.5275% and the average Kappa value reached 0.8340.

Figure 4 is a histogram representation of the accuracy of the 9 subjects, Figure 5 is a histogram representation of the Kappa value of the 9 subjects:

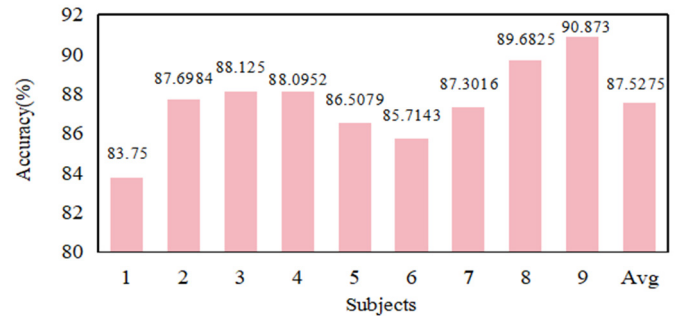


Figure 4 Accuracy histogram

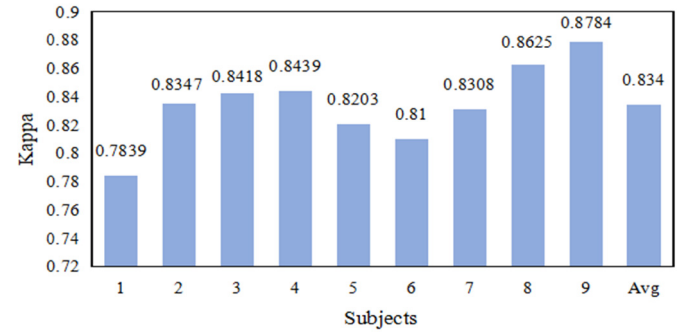


Figure 5 Kappa histogram

Tables 2 and 3 present the findings of the several groups of researchers in this study area. Table 2 shows the accuracy results of the four groups of studies on the BCI Competition 2a dataset. Yimin Hou et al. used three methods of bispectrum, entropy and common space pattern in higher order spectra to extract multiple EEG signal features and classified them using SVM, proposed Bispectrum, Entropy and common spatial pattern (BECSP) algorithm and compared the classification accuracy of traditional CSP feature extraction algorithm. The average accuracy of the BECSP algorithm on the BCI Competition 2a dataset with 9 subjects was 71.61%, which is a 2.05% improvement over the traditional CSP algorithm. The average classification accuracy of the method proposed in this study is 15.9175% higher than BECSP.

Bao Liu et al. used discrete wavelet transform for data denoising, and used convolutional neural network (CNN) and temporal convolutional network (TCN) to extract spatio-temporal features and classify them. Also the study compared the accuracy of classification using only CNNs. The average accuracy of their proposed method on 9 subjects of BCI Competition 2a dataset is 85.53%, which is 8.64% better than the CNN method. The average classification accuracy of the proposed method in this study is 1.9975% higher than that of the method.

Also this study compares two traditional feature extraction methods Filter Bank Common Spatial Pattern (FBCSP) and Common Spatio-spectral Pattern (CSSP). The accuracy of the proposed method in this study is improved by 24.3575% and 9.1275% over these two methods respectively.

The classification accuracy of the methods in literature 9 and literature 10 varies widely across subjects with, the highest

differences of 40.97% and 20.49%, respectively, and the generalization ability was weak. The method proposed in this study has a maximum difference in classification accuracy of 7.123% across subjects, is weakly influenced by the subjects, and has a high generalization ability.

Table 2 Recognition accuracy of different algorithms for 9 subjects

Subject	Acc (%)					
	CSP[9]	BECSP [9]	CNN [10]	E-EEG [10]	FBCSP [11]	CSSP [12]
1	78.47	80.9	78.47	89.93	77.4	90.97
2	61.45	64.24	63.89	77.08	54.2	56.94
3	81.94	85.76	85.42	96.18	69.8	92.01
4	62.84	65.63	80.56	81.25	56.3	72.92
5	40.97	44.79	77.43	80.21	46.9	61.81
6	52.77	54.17	67.36	75.69	52.1	65.28
7	83.68	84.38	81.25	93.40	83.0	79.86
8	82.98	84.03	80.56	88.19	60.4	93.06
9	80.9	80.56	77.08	87.85	68.4	92.36
Ave	69.56	71.61	76.89	85.53	63.17	78.40

Table 3 shows the Kappa results of the two groups of studies on the BCI Competition 2a dataset. The average Kappa of the BECSP algorithm on the BCI Competition 2a dataset with 9 subjects was 0.621. The average Kappa of the proposed method in this study was improved by 0.213 over the BECSP method.

The average Kappa of the method proposed by Bao Liu et al. on 9 subjects of BCI Competition 2a dataset is 0.8071, which is 0.1152 higher than the CNN method. The average Kappa of the method proposed in this study was improved by 0.0269 over that method.

The Kappa of the above two methods on different subjects differed significantly, with the highest differences of 0.546 and 0.2732, respectively, and the generalization ability was weak. The method proposed in this study has the highest difference in classification accuracy of 0.0945 on different subjects, which is weakly influenced by the subjects and has a strong generalization ability.

Table 3 Recognition kappa of different algorithms for 9 subjects

Subject	Kappa		
	BECSP[9]	CNN[10]	E-EEG[10]
1	0.745	0.7130	0.8657
2	0.523	0.5185	0.6944
3	0.81	0.8056	0.9491
4	0.542	0.7407	0.7500
5	0.264	0.6991	0.7361
6	0.389	0.5648	0.6759
7	0.792	0.7500	0.9120
8	0.787	0.7407	0.8426
9	0.741	0.6944	0.8380
Ave	0.621	0.6919	0.8071

IV. CONCLUSIONS

In this study, a feature extraction model named CCWT was proposed for motor imagery EEG signal feature extraction. The extracted features contain information in both the time-

frequency and space domains using a frequency band filter bank co-space mode as well as a continuous wavelet transform. For classification and recognition, this study uses Convolutional Neural Networks combined with Long Short-Term Memory (LSTM) for classification. The method proposed in this study obtained 87.5275% accuracy as well as 0.8340 Kappa value on BCI Competition 2a dataset, obtained good classification results and provided more possibilities for motor imagery.

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